



WORLD HEAT FLOW DATABASE PROJECT

The World Heat Flow Database Project (2022-2025) aimed to develop a new research data infrastructure for the IHFC Global Heat Flow Database that offers a one-stop shop for comprehensive information on heat-flow-related data, publications, projects, and researchers. Supported by the International Heat Flow Commission ([IHFC](#)), this new user-oriented web service was designed to the geoscientific community quality-proofed, up-to-date, well-documented, extended, enriched, and restructured heat-flow data. The project supports the [Global Heat Flow Data Assessment Project](#), led by IHFC and Task Force VIII of the International Lithosphere Program ([ILP](#)).

The new technical database reflects the criteria of FAIR and OPEN data policy and supports the interoperability with other geoscientific data services (e.g. [EPOS](#)). Core elements are the provision of globally used persistent identifiers: (i) [DOI](#) for heat-flow data newly submitted to the database, (ii) [ORCID](#) IDs to uniquely identify authors, and (iii) International Generic Sample Numbers ([IGSN](#)) for connecting data and literature with physical samples on which data were measured.

The project (phase one: 04.2022-06.2025) was funded by the "The [Deutsche Forschungsgemeinschaft](#)" (DFG, German Research Foundation). Project partners were the GFZ Helmholtz Centre for Geosciences and the Technical University of Dresden. [Contact](#) the project PI. The team behind the development of phase one is listed [here](#).

Project record

After the launch of the portal, the project phase one was completed.

- July 2025 - Portal launching event (1/2 days, Potsdam hybrid)
- May 2025 - Final project workshop (2 days, Potsdam)
- Apr 2025 - FOSSGIS 2025 - [\[presentation\]](#)
- Feb 2025 - 12th project workshop (1 days online)
- Dec 2024 - 11th project workshop (2 days, Potsdam)
- Jun 2024 - 10th project workshop (2 days, Dresden)
- Apr 2024 - presentations and project stakeholder workshop as splinter meeting at EGU2024 in Vienna, Austria [[heatflow spring 2024](#)]
- Mar 2024 - FOSSGIS 2024 - [\[presentation\]](#)
- Feb 2024 - 7th project workshop (2 days, Potsdam)
- Mar 2024 - Community and stakeholder kick-off meeting
- Oct 2023 - 6th project workshop (19-20 Oct, Dresden)
- Aug 2023 - 5th project workshop (27-28 Aug, Potsdam)
- July 2023 - IUGG summer school and workshops (3-11 July, Berlin) [[heatflow summer 2023](#)]
- May 2023 - 4th project workshop (Potsdam)
- Mar 2023 - 3rd project workshop (2-3 March, Potsdam)
- Dec 2022 - 2nd project workshop (1-2 December, Dresden)
- Jun 2022 - 1st project workshop (30-1 July, Potsdam)
- Jun 2022 - Cermak7 Meeting (20-22 June, Potsdam) [\[conference page\]](#)
- May 2022 - Kick-off meeting

Publications and Outreach

Paper

- Florian Neumann, Ben Norden, Elif Balkan-Pazvantoğlu, Samah Elbarbary, Alexey G. Petrunin, Kirsten Elger, Samuel Jennings, Simone Frenzel, and Sven Fuchs (PREPRINT) The 2024 Release of the Global Heat Flow Database (GHFDB): Quality Assessment, Metadata Standards, and a Century of Geothermal Data. **Earth System Science Data**. <https://essd.copernicus.org/preprints/essd-2025-341/#discussion>
- Fuchs, S., Norden, B., Neumann, F., et al. (2023): Quality-assurance of heat-flow data: The new structure and evaluation scheme of the IHFC Global Heat Flow Database, **Tectonophysics**, <https://doi.org/10.1016/j.tecto.2023.229976>
- Ott, N., & Mäs, S. (2023): Analysis of free web mapping libraries. **Zenodo**. <https://doi.org/10.5281/zenodo.8139086>.
- Fuchs, S., Förster, A., Norden, B. (2022): Evaluation of the terrestrial heat flow in Germany: A case study for the reassessment of global continental heat-flow data. **Earth-Science Reviews**, Volume 235, December 2022, 104231, <https://doi.org/10.1016/j.earscirev.2022.104231>

Code & Templates

- Fuchs, Sven; Neumann, Florian; Balkan-Pazvantoğlu, Elif; Elbarbary, Samah; Petrunin, Alexey G.; Jennings, Samuel; Elger, Kirsten; Norden, Ben (2025): Global Heat Flow Database Data Template. V. 2025.06. **GFZ Data Services**. <https://doi.org/10.5880/fidgeo.2025.042>
- Chishti, Saman F.; Balkan-Pazvantoğlu, Elif; Norden, Ben; Neumann, Florian; Elbarbary, Samah; Gross, Eskil S.; Petrunin, Alexey G.; Fuchs, Sven (2025): Heat Flow Quality Analysis Toolbox (hfqa_tool). GFZ Data Services. doi.org/10.5880/fidgeo.2025.043

Data

- Global Heat Flow Data Assessment Group; Fuchs, Sven; Neumann, Florian; Norden, Ben; Balkan-Pazvantoglu, Elif; Elbarbary, Samah; Petrunin, Alexey; Beardsmore, Graeme; Harris, Robert; Negrete-Aranda, Raquel; Poort, Jeffrey; Verdoya, Massimo; Liu, Shaowen; Chambers, Emma; Fuentes-Bustillos, Karina; Sidagam, Eswara Rao; Matiz-Leon, Jhon Camilo; Bencharef, Mohammed Hichem; Mino, Belay G.; Khaled, Mohamed Shafik; Verch, Denise; Berger, Leonard; Chishti, Saman Firdaus; Dergunova, Viktoria; Liebing, Helena; Schulz, Marvin; Schuppe, Pia; Trepalova, Zlata; Chiozzi, Paolo; Duque, Maria Rosa Alves; Forster, Florian; Leveni, Martina; Staal, Tobias (2024): The Global Heat Flow Database: Release 2024. **GFZ Data Services**. <https://doi.org/10.5880/fidgeo.2024.014>
- Fuchs, S., Neumann, F., Norden, B., Beardsmore, G., Chiozzi, P., Colgan, W., Anguiano Dominguez, A. P., Duque, M. R. A., Ojeda Espinoza, O. M., Forster, F., Förster, A., Fröhder, R., Fuentes-Bustillos, K., Hajto, M., Harris, R., Jöeleht, A., Liebing, H., Liu, S., Lüdtk, G., Madon, M., Negrete-Aranda, R., Poort, J., Reznik, I. J., Riedel, M., Rolandone, F., Stål, T., Verdoya, M., Wu, J.-N. (2023): The Global Heat Flow Database: Update 2023. **GFZ Data Service**. <https://doi.org/10.5880/fidgeo.2023.008>

Conferences and presentations

- Ott, N., Elger, K., Frenzel, S., Fuchs, S., Jennings, S., Mäs, S., Neumann, F., Norden, B. (2025): Einbindung von OGC API für Wärmestromdaten. **FOSSGIS 2025**: Münster, Germany. Zenodo. <https://doi.org/10.5281/ZENODO.10571906>
- Ott, N., Mäs, S., Fuchs, S., Elger, K., Jennings, S., Neumann, F., Norden, B., & Frenzel, S. (2024): Weltwärmestrom Datenbank Projekt: webbasierte Explorationswerkzeuge für Punktdaten. **FOSSGIS 2024**: Hamburg, Germany. Zenodo. <https://doi.org/10.5281/ZENODO.10571906>
- Neumann, F., Fuchs, S., Norden, B., Balkan-Pazvantoğlu, E., Petrunin, A., Elbarbary, S., Jennings, S., Elger, K., Frenzel, S., Ott, N., Mäs, S., & GHFD Assessment Group (2024): The heat flow data assessment project: Transformation of the global heat flow database. **EGU General Assembly 2024**, Vienna, Austria, <https://doi.org/10.5194/egusphere-egu24-16952>
- Jennings, S., Elger, K., Fuchs, S., Neumann, F., Norden, B., Frenzel, S., Mäs, S., & Ott, N. (2024): Geoluminate: A community-centric framework for the creation, deployment and ongoing development of decentralized geoscience data portals. **EGU General Assembly 2024**, Vienna, Austria, <https://doi.org/10.5194/egusphere-egu24-19849>
- Mäs, S., Elger, K., Fuchs, S., Ott, N., Jennings, S., Norden, B., Neumann, F., Frenzel, S. (2023): Heatflow.world: an international research data infrastructure for the Global Heat Flow Database. **NFDI4Earth - Interessensgruppe Geologie & Geophysik (I3G) Workshop** 29.11.2023
- Norden, B., Förster, A., Fuchs, S. (2023): The quality assessment of the German heat flow database, XXVIII **General Assembly of the International Union of Geodesy and Geophysics (IUGG)** (Berlin 2023). <https://doi.org/10.57757/IUGG23-3270>
- Fuchs, S., Förster, A., Norden, B. (2023) A new quality assessed heat flow database for Germany, **DGG Jahrestagung 2022** Bremen, Germany, 06.03.-09.03.2023 (ORAL)
- Neumann, F.; Norden, B.; Fuchs, S. (2022) The Global Heat Flow Data Assessment Project and Fellowship Program **RAUGM 2022, Mexico**, 30.10.-04.11.2022 (POSTER)
- Fuchs, S., The Global Heat Flow Data Assessment Group (2022): The global heat flow database - status, progress and future projects, **Cermak7**: 7th International Meeting on Heat Flow and the Thermal Structure of the Lithosphere, Potsdam, Germany, 20-22.06.2022 (ORAL)
- Norden, B., Förster, A., Fuchs, S. (2022) Quality assessment of German Heat Flow data: a contribution to a revised Global Heat Flow Database, **Cermak7**: 7th International Meeting on Heat Flow and the Thermal Structure of the Lithosphere, Potsdam, Germany, 20-22.06.2022 (POSTER) ([conference page](#))



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